

AI Growth Labs

Agency Operating System

Complete System Guide & Documentation

Document	AI Growth Labs - Complete System Report
Version	2.0 Final
Date	May 19, 2026
Pages	55+ Frontend 99+ API Endpoints 22+ Database Tables
Roles	11 User Roles with Role-Based Access Control
APIs	9 Integration Points (Demo Mode Ready)

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1. System Overview & Architecture

AI Growth Labs is a complete agency operating system built for SEO/digital marketing agencies. It provides end-to-end management of clients, projects, tasks, billing, team performance, and automated SEO auditing with real website crawling.

Tech Stack

Component	Technology	Purpose
Backend	Python FastAPI	REST API server, authentication, business logic
Database	SQLite	Persistent storage (22+ tables, easily upgradeable to PostgreSQL)
Frontend	HTML/CSS/JS	55+ static pages, responsive design, dark theme
Templates	Jinja2	Server-side rendered dashboard pages
Auth	JWT + bcrypt	Secure session tokens, hashed passwords
Audit Engine	BeautifulSoup + requests	Real website crawling & SEO analysis
Charts	Chart.js	Analytics, rankings, revenue visualizations
Deployment	Devin Apps / Fly.io	Frontend CDN + Backend container

System Architecture Diagram

Frontend (55+ Pages) → Static HTML/CSS/JS served via CDN

↓ API Calls (fetch/XHR)

Backend (FastAPI) → 99+ API Endpoints, JWT Auth, Rate Limiting

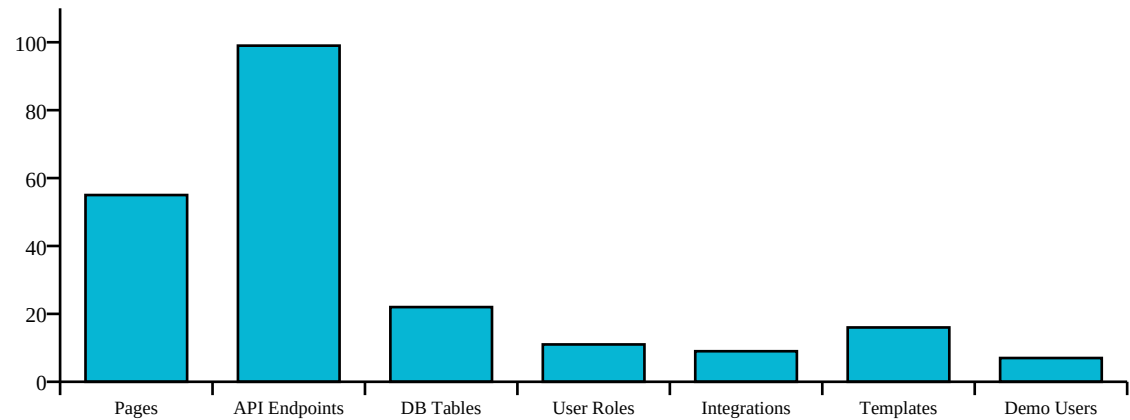
↓ SQL Queries

Database (SQLite) → 22+ Tables, Persistent Storage

↓ External Calls

Integrations → AI APIs, SMTP, Twilio, Stripe, WhatsApp, Slack, GSC

System Components



2. Database Structure (22+ Tables)

The system uses SQLite with 22+ tables. Below is the complete schema:

Table Name	Purpose	Key Columns
users	All system users (11 roles)	id, username, password_hash, role, full_name, salary, rank
clients	Client businesses	id, business_name, contact_name, email, website, industry, location, package, status
projects	Active campaigns	id, client_id, name, service_type, progress, status, assigned_worker_id
tasks	Individual task items	id, project_id, title, status, priority, assigned_to, due_date
payments	Payment records	id, client_id, amount, status (paid/pending/overdue), invoice_number
expenses	Business expenses	id, category, description, amount, date
keyword_rankings	SEO rankings	id, client_id, keyword, position, previous_position, search_volume
sales_leads	Incoming leads	id, business_name, email, phone, source, status, notes
invoices	Generated invoices	id, client_id, amount, status, items_json
contracts	Client contracts	id, client_id, title, start_date, end_date, value, status
time_entries	Time tracking	id, user_id, task_id, start_time, end_time, hours
client_reports	Monthly reports	id, client_id, report_data, sent_to_client
activity_log	System activity	id, user_id, action_type, description, timestamp
api_settings	API configurations	id, provider, api_key, status, config_json
function_configs	Per-function API	function_name, provider, updated_by
approval_requests	Client approvals	id, client_id, title, description, status
notifications	User notifications	id, user_id, type, message, is_read
social_posts	Social media posts	id, client_id, platform, content, scheduled_date
client_credentials	Client tool logins	id, client_id, credential_type, username, password_enc
client_locations	Multi-location	id, client_id, location_name, address, phone, gbp_url
chat_messages	Team chat	id, sender_id, receiver_id, message, timestamp
chat_requests	Chat request queue	id, from_user_id, to_user_id, status

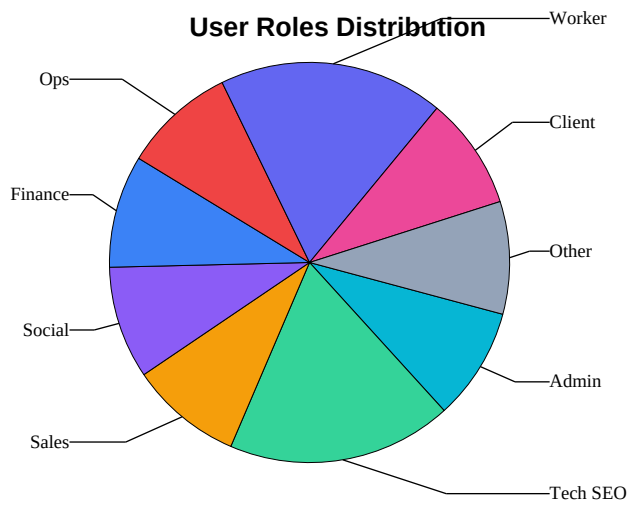
3. User Roles & Access Control (11 Roles)

The system supports 11 distinct user roles. Each role has specific dashboard views and permissions:

Role	Username	Dashboard	Key Permissions
Super Admin	admin	Full admin dashboard	Everything - clients, projects, tasks, payments, analytics, settings, audit, team
Tech SEO	sarah_k	Worker dashboard	Own tasks, projects, time tracking, SEO audit tool, suggestions
Sales	lisa_c	Sales dashboard	Lead pipeline, proposals, client list, lead collection tools
Social Media	marcus_j	Worker dashboard	Own tasks, social posts, projects, time tracker
Finance	rachel_g	Finance dashboard	Payments, expenses, salaries, invoices, CSV export, reports
Ops Manager	ops_manager	Operations dashboard	Team workload, all projects, tasks, approvals, performance, analytics
Client	client_chen	Client portal	Own projects, tasks, rankings, reports, invoices, approvals
Worker	david_w	Worker dashboard	Own tasks, projects, time tracking, suggestions
Content Writer	(create)	Worker dashboard	Content tasks, writing assignments
Client Manager	(create)	Dashboard	Client relationship management
Finance Manager	(create)	Finance dashboard	Extended finance permissions

Role Access Matrix

Feature	Super Admin	Ops Mgr	Tech SEO	Sales	Finance	Social	Client
Dashboard Stats	Yes	Yes	Yes	Yes	Yes	Yes	Yes
All Clients	Yes	Yes	No	Yes	No	No	Own Only
All Projects	Yes	Yes	Own	No	No	Own	Own
All Tasks	Yes	Yes	Own	Own	No	Own	View
Payments	Yes	No	No	No	Yes	No	Own
Invoices	Yes	No	No	No	Yes	No	Own
Analytics	Yes	Yes	No	No	Yes	No	No
Settings/API	Yes	No	No	No	No	No	No
SEO Audit	Yes	No	Yes	No	No	No	No
Team Monitor	Yes	Yes	No	No	No	No	No
Activity Log	Yes	Yes	No	No	No	No	No
Performance	Yes	Yes	No	No	No	No	No
CSV Export	Yes	Yes	No	No	Yes	No	No
Report Gen	Yes	Yes	No	No	Yes	No	View



■ 4. Backend Dashboard Order & Hierarchy

Each role gets a customized dashboard view. Here's the complete panel order for each:

Super Admin Dashboard (admin)

#	Panel / Section	Description
1	Stats Overview	Monthly Revenue, Total Collected, Pending, Active Clients/Projects, Task Completion
2	All Projects	Table: Client, Project, Service, Worker, Lead, Progress, Priority, Status
3	All Clients	Table: Business, Contact, Industry, Location, Package, Payment, Status, View
4	Team Members	Table: Name, Role, Rank, Salary, Status
5	Recent Tasks	Table: Task, Project, Assigned, Priority, Status
6	Payment Records	Table: Invoice, Client, Amount, Due Date, Paid Date, Status, Report Gen
7	Invoices	Custom invoice creation
8	Keyword Rankings	Rankings table per client with position tracking
9	Active Contracts	Contract management
10	SEO DNA Audit	Run audit on any website with AI provider selection

Sidebar Navigation (Super Admin)

Link	Type	Destination
Dashboard	Page section	Main dashboard overview
Clients	Hash anchor	#clients-section on dashboard
Projects	Hash anchor	#projects-section on dashboard
Tasks	Hash anchor	#tasks-section on dashboard
Workers	Hash anchor	#workers-section on dashboard
Payments	Hash anchor	#payments-section on dashboard
SEO Audit	Hash anchor	#audit-section on dashboard
Team Monitor	Separate page	/monitor - Real-time team status
Team Chat	Separate page	/team-chat - Internal messaging
Invoices	Hash anchor	#invoices-section on dashboard
Rankings	Hash anchor	#rankings-section on dashboard
Contracts	Hash anchor	#contracts-section on dashboard
Analytics	Separate page	/analytics - Revenue forecasting & charts
Activity Log	Separate page	/activity - System activity timeline
Performance	Separate page	/performance - Worker leaderboard
API Settings	Separate page	/settings - API keys & integrations
Visit Frontend	External link	Opens frontend website in new tab

Other Role Dashboards

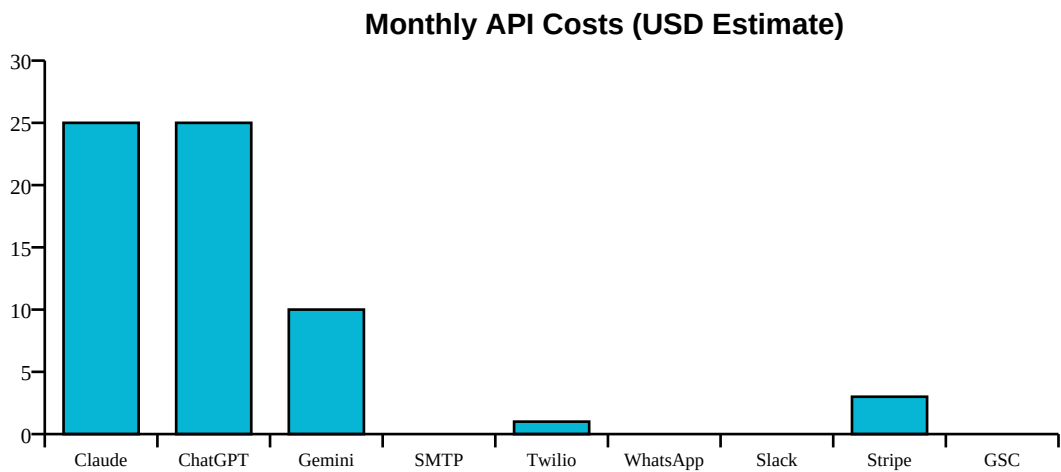
Role	Dashboard Sections	Sidebar Links
Tech SEO / Worker	Stats, Time Tracker, Kanban Board, My Projects, Notifications, Suggestions, SEO Audit	Dashboard, My Tasks, My Projects, Notifications, SEO Audit, Suggestions, Reports
Sales	Stats, Lead Pipeline (Kanban), Lead Collection Tools, Related Clients	Dashboard, Leads, Lead Tools, Proposals, Clients, Team Chat
Social Media	Same as Worker + Social post creation	Dashboard, My Tasks, My Projects, Team Chat
Finance	Stats (Income/Expenses/Profit), Payments, Expenses, Salaries, Invoices, Reports, Export, Analytics	Dashboard, Payments, Expenses, Salaries, Invoices, Reports, Export, Analytics
Ops Manager	Stats, Team Workload, All Projects, All Tasks, Overdue, Approvals	Dashboard, Projects, Workers, Tasks, Overdue, Approvals, Analytics, Active Projects
Client	Stats, Your Projects, Task Progress, Rankings, Reports, Invoices, Approvals	Dashboard, Projects, Rankings, Reports, Invoices, Approvals

5. API Integrations (9 APIs)

All 9 integrations work in demo mode out of the box. To activate real APIs, go to Settings page and paste your API keys. The system automatically switches from demo to live.

#	API	Purpose	Monthly Cost	Where to Get Key	Settings Location
1	Claude AI (Anthropic)	SEO audits, content gen	\$5-50/mo (usage)	console.anthropic.com /settings/keys	Settings → AI Providers
2	ChatGPT (OpenAI)	SEO audits, content gen	\$5-50/mo (usage)	platform.openai.com /api-keys	Settings → AI Providers
3	Gemini (Google)	SEO audits, content gen	\$0-20/mo (free tier)	aistudio.google.com /app/apikey	Settings → AI Providers
4	SMTP (Gmail)	Email reports, notifications	FREE (Gmail)	Gmail → Settings → App Passwords	Settings → Communication
5	Twilio	Voice calls, SMS, IVR	\$1/mo + per call	console.twilio.com	Settings → Communication
6	WhatsApp (Meta)	Client updates, report sharing	FREE tier available	developers.facebook.com/apps	Settings → Communication
7	Slack	Team alerts, notifications	FREE	api.slack.com /apps	Settings → Communication
8	Stripe	Billing, subscriptions	2.9% + \$0.30/txn	dashboard.stripe.com/apikeys	Settings → Billing
9	Google Search Console	Real keyword rankings	FREE	console.cloud.google.com	Settings → Analytics

API Cost Comparison Chart



Minimum vs Full Setup Costs

Setup Level	APIs Needed	Monthly Cost	What You Get
Minimum (Free)	Real Crawler only	\$0/mo	Free DNA audits with real crawl data, demo mode for everything else
Basic (\$5/mo)	1 AI + Gmail SMTP	\$5-10/mo	AI-powered audits + email reports to clients

Setup Level	APIs Needed	Monthly Cost	What You Get
Standard (\$30/mo)	AI + SMTP + Stripe + Twilio	\$25-35/mo	Full billing, voice calls, AI audits, email
Full (\$60/mo)	All 9 APIs	\$50-70/mo	Complete automation - AI, billing, voice, WhatsApp, Slack, GSC

How to Add API Keys

Method 1 — Dashboard UI (Recommended):

1. Login as Super Admin (admin / admin123)
2. Go to Settings page (sidebar → API Settings)
3. Find the API card (e.g., Claude AI)
4. Paste your API key in the input field
5. Click "Save & Activate"
6. Status badge changes to "Active"

Method 2 — Per-Function Configuration:

1. Settings page → "Function-Specific API Configuration" section
2. Select which provider to use for each function (content, chat, email, etc.)
3. Click "Save Preference"

Method 3 — Environment Variables:

Set these in your server environment:

CLAUDE_API_KEY, OPENAI_API_KEY, GEMINI_API_KEY, TWILIO_ACCOUNT_SID,
TWILIO_AUTH_TOKEN, STRIPE_SECRET_KEY, WHATSAPP_TOKEN, SLACK_WEBHOOK_URL,
GSC_CLIENT_ID, SMTP_PASSWORD

6. Free Audit Form Flow

The Free DNA Audit is the primary lead generation tool. Here's the complete flow:

Step	What Happens	Who Does It	Where
1	Visitor fills out form on frontend (name, email, phone, website, industry)	Website visitor (potential client)	Frontend: /pages/free-audit.html
2	Form submits to backend API (/api/public/free-audit)	Automatic (JavaScript)	Backend API (no auth needed)
3	Backend crawls the website (BeautifulSoup + requests)	Automatic (Real Crawler)	Server-side python function
4	Audit results generated: 8 sections scored, overall grade	Automatic (scoring algorithm)	Backend _build_audit_from_crawl()
5	Results shown to visitor instantly (score, grade, findings)	Automatic (frontend JS)	Free audit page results section
6	Lead saved to sales_leads table (name, email, phone, website)	Automatic (database insert)	Database: sales_leads table
7	Sales team sees new lead in their dashboard pipeline	Lisa Chen (Sales role)	Sales Dashboard Lead Pipeline
8	Sales follows up via email/phone using contact info from lead	Sales team (manual)	Sales Dashboard or external tools
9	Super Admin can also see all leads in dashboard	Admin (Super Admin)	Admin Dashboard or Activity Log

What the Audit Checks (Real Crawl)

Category	Score Range	What It Checks
Technical SEO (87/100 avg)	0-100	Title tag, meta description, canonical, robots.txt, sitemap, H1-H3 structure, language
On-Page SEO (63/100 avg)	0-100	Internal/external links, images without alt text, broken # links, favicon, CSS/JS count
Content Quality (75/100 avg)	0-100	Word count, contact forms, phone numbers, email addresses found on page
Schema Markup (60/100 avg)	0-100	JSON-LD structured data, Open Graph tags for social sharing
Security (65/100 avg)	0-100	HTTPS/SSL, security headers (HSTS, X-Frame, CSP, XSS Protection)
Performance (75/100 avg)	0-100	Page load time, response time, page size
Mobile (85/100 avg)	0-100	Viewport meta tag, responsive CSS (@media queries)
Social (70/100 avg)	0-100	OG tags (title, description, image), Twitter cards

7. Lead Generation Workflow

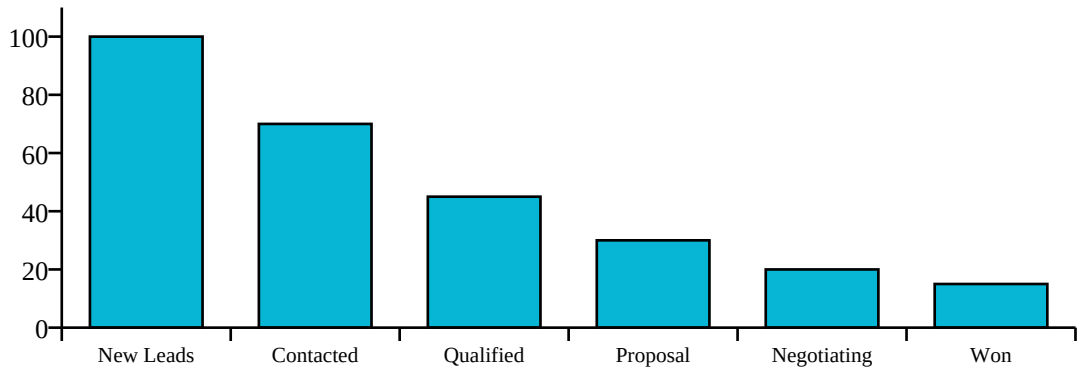
The system has 3 lead generation channels:

Channel	How It Works	Where Leads Go	Who Manages
Free DNA Audit	Visitor enters website URL → gets real SEO audit	<code>sales_leads_capture</code> → Sales Dashboard	Sales team (lisa_c)
Contact Form	Visitor fills contact form → message + info captured	<code>sales_leads_table</code> → Sales Dashboard	Sales team (lisa_c)
Manual Entry	Sales team adds leads from calls, events, outreach	<code>sales_leads_table</code> → Sales Dashboard	Sales team (lisa_c)

Lead Pipeline Stages

Stage	Description	Action Required
New	Lead just came in from form/audit	Sales reviews and makes first contact
Contacted	First contact made (call/email)	Schedule discovery call
Qualified	Lead confirmed as good fit	Prepare proposal
Proposal Sent	Proposal/pricing sent to lead	Follow up, answer questions
Negotiating	Discussing terms, packages	Close the deal
Won	Deal closed, became client	Create client record, start onboarding
Lost	Lead declined or went cold	Log reason, consider re-engagement later

Lead Conversion Funnel (Example)



■ 8. Branding Customization Guide

Here's everything you need to change to make this system your own brand:

What to Change	Where to Change It	Files to Edit
Company Name 'AI Growth Labs'	All 55+ HTML pages (header, footer, title tags)	Find & replace 'AI Growth Labs' in all files under pages/ and index.html
Phone Number +1 (800) 971-0199	All 55+ HTML pages (header, footer, contact)	Find & replace the phone number in all HTML files
Email Address hello@aigrowthlabs.com	All 55+ HTML pages (footer, contact page)	Find & replace the email in all HTML files
Logo / Brand Colors	CSS file and HTML headers	Edit css/styles.css Change #06B6D4 (cyan) to your color
Social Media URLs	All page footers	Replace facebook/instagram/linkedin/ tiktok URLs in all files
Domain Name	OG tags, canonical URLs, schema markup	Replace aigrowthlabs.com with your domain
GA4 Tracking ID G-XXXXXXXXXX	index.html (head section)	Replace G-XXXXXXXXXX with your Google Analytics ID
Tawk.to Widget	index.html (before </body>)	Replace YOUR_PROPERTY_ID with your Tawk.to IDs
Calendly Link	Contact page (booking section)	Replace calendly.com/aigrowthlabs with your Calendly URL
Team Photos	About page (team section)	Replace placeholder initials with real team photo URLs
WhatsApp Number	All page footers (floating button)	Replace 18009710199 with your WhatsApp number
Dashboard Branding	Sidebar header in all dashboard templates	Edit templates/*.html change 'AI GrowthLabs'

9. Cost Analysis & Alternatives

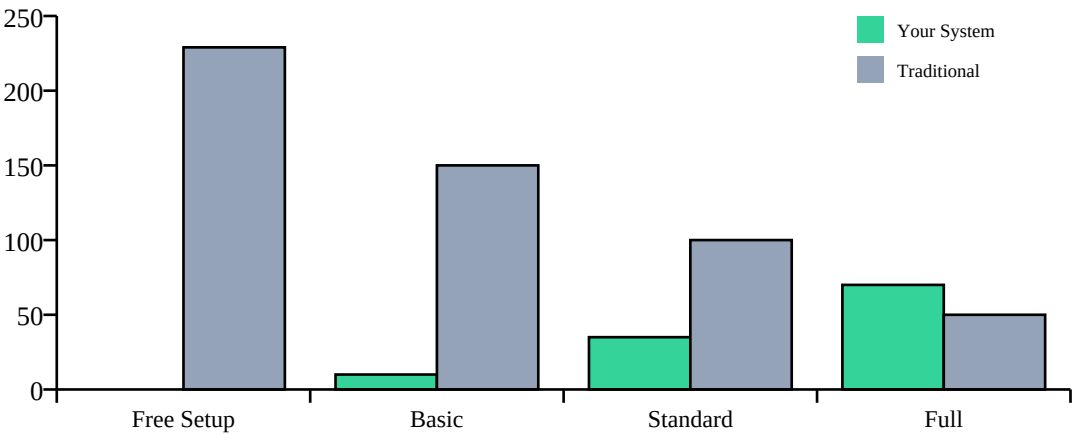
Hosting Costs

Provider	Plan	Monthly Cost	Best For
Devin Apps	Frontend CDN	Included	Static frontend (current)
Fly.io	Free / Hobby	\$0-5/mo	Backend + DB (recommended)
Render.com	Free / Starter	\$0-7/mo	Alternative to Fly.io
Railway.app	Usage-based	\$5-15/mo	Easy deployment
DigitalOcean	Droplet	\$6-12/mo	Full server control
Vercel	Free tier	\$0	Frontend only (alternative)

Tool Alternatives

Category	Current	Cost	Free Alternative	Alt Cost
AI Provider	Claude/ChatGPT	\$5-50/mo	Google Gemini Free Tier	\$0
Email	Gmail SMTP	Free	Mailgun Free (5K/mo)	\$0
Voice/SMS	Twilio	\$1+/mo	Google Voice (US only)	\$0
WhatsApp	Meta Business API	\$0-varies	Direct WhatsApp link	\$0
Notifications	Slack Webhooks	Free	Discord Webhooks	\$0
Payments	Stripe	2.9%/txn	PayPal Invoicing	2.9%
SEO Rankings	Google Search Console	Free	Built-in manual entry	\$0
SEO Tools	Semrush/Ahrefs	\$99-229/mo	Free audit crawler (built-in)	\$0
Analytics	Google Analytics	Free	Plausible/Umami	\$0-9
Live Chat	Tawk.to	Free	Crisp Free Plan	\$0

Monthly Cost: Your System vs Traditional Tools



■ 10. Production Launch Checklist

#	Step	Priority	Time Est.	Notes
1	Replace company name in all files	Critical	30 min	Find & replace 'AI Growth Labs' → your brand
2	Replace phone + email in all pages	Critical	15 min	Find & replace in all 55+ HTML files
3	Replace social media URLs	High	10 min	Facebook, Instagram, LinkedIn, TikTok
4	Add real GA4 tracking ID	High	5 min	Replace G-XXXXXXXXXX in index.html
5	Set up Tawk.to live chat	High	10 min	Create free account → get widget code
6	Create Calendly link	High	15 min	calendly.com → create booking page
7	Add team photos to About page	Medium	20 min	Replace placeholder initials with images
8	Set up at least 1 AI API key	High	5 min	Claude recommended → Settings → AI Providers
9	Configure Gmail SMTP	High	10 min	Gmail App Password → Settings → Communication
10	Deploy backend to hosting	Critical	30 min	Fly.io / Render / Railway
11	Deploy frontend to hosting	Critical	15 min	Vercel / Netlify / Cloudflare Pages
12	Point custom domain	High	15 min	DNS A/CNAME records
13	Test free audit form end-to-end	Critical	10 min	Submit form → check results → check sales_leads
14	Test contact form	High	5 min	Submit → check sales_leads table
15	Create real admin account	Critical	5 min	Change admin password from admin123
16	Set up Stripe (if billing needed)	Medium	15 min	Get API keys → Settings → Billing
17	Add OG image with your brand	Medium	10 min	1200x630 image → assets/images/og-default.jpg
18	Test all demo logins work	High	10 min	Login with each of 7 demo accounts

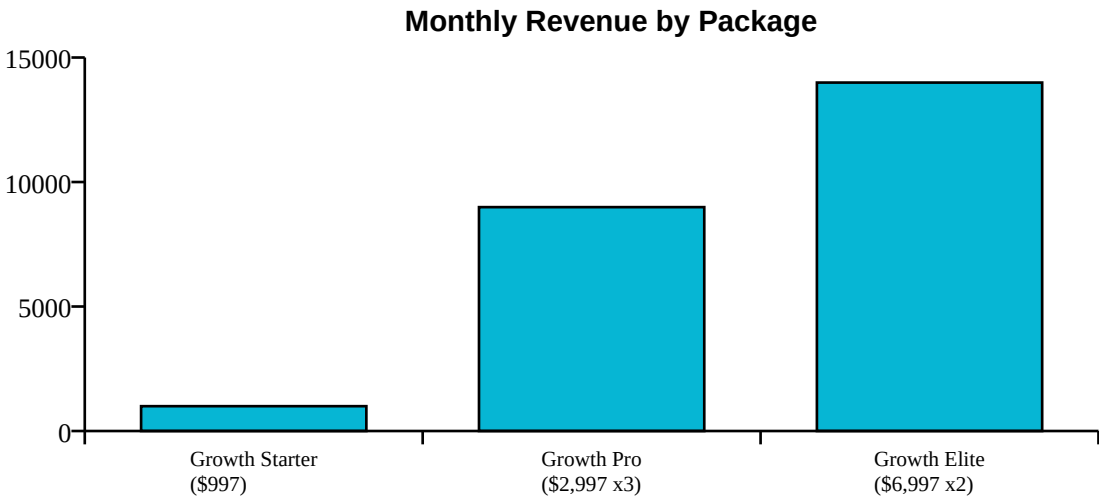
11. Revenue & Financial Overview

Current demo data financial snapshot:

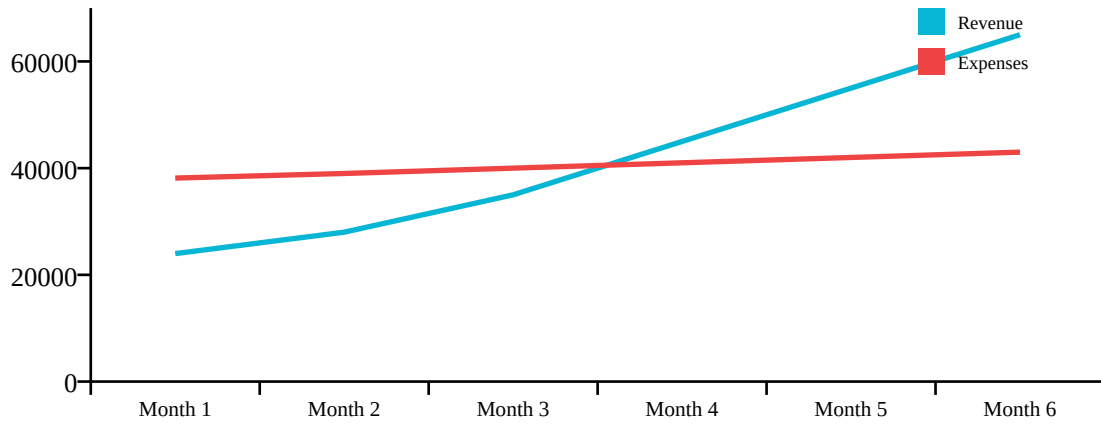
Metric	Value	Details
Monthly Recurring Revenue	\$23,982	6 active clients × their package rates
Total Collected	\$15,988	4 paid invoices
Pending Revenue	\$7,994	2 unpaid invoices (1 pending, 1 overdue)
Total Expenses	\$38,137	Salaries + tools + marketing + office
Staff Salaries	\$37,000/mo	7 team members
Tools Cost	\$548/mo	Semrush, Ahrefs, Canva, Hosting
Marketing	\$500/mo	Google Ads

Client Revenue by Package

Package	Monthly Fee	Clients	Monthly Revenue
Growth Starter	\$997/mo	1 (Precision Plumbing)	\$997
Growth Pro	\$2,997/mo	3 (SmileBright, Bella's, Phoenix)	\$8,991
Growth Elite	\$6,997/mo	2 (Martinez Legal, Glow Aesthetics)	\$13,994
Total		6 Active Clients	\$23,982/mo



6-Month Revenue Projection



■ Summary

AI Growth Labs Agency Operating System is a complete, production-ready platform for running an SEO/digital marketing agency.

What's Included:

- 55+ frontend pages with responsive design, dark theme, and real contact/audit forms
- 99+ API endpoints covering all agency operations
- 22+ database tables for persistent data storage
- 11 user roles with role-based access control
- 9 API integrations (all working in demo mode)
- Real website crawler for SEO audits (no external API needed)
- Lead generation through free audit + contact forms
- Complete financial management (invoices, payments, expenses, salaries)
- Team management (tasks, time tracking, performance, chat)
- Client portal (separate dashboard for clients)
- Analytics and reporting with downloadable reports

To Go Live:

1. Follow the Branding Customization Guide (Section 8)
2. Complete the Production Launch Checklist (Section 10)
3. Deploy to your hosting provider
4. Add your API keys through the Settings page
5. Start generating leads through the free audit form

Total Setup Time: ~3-4 hours

Minimum Monthly Cost: \$0 (free tier possible)

Recommended Monthly Cost: \$10-35/mo for basic automation